



Abstract

Identifying genes associated with diseases is crucial to understanding disease mechanisms and developing therapies, but individual gene-level findings must be supplemented with clustering analysis to reveal gene modules and their relationships. In this work, we perform robust clustering analysis on RNA-Seq data of **Age-related Macular Degeneration (AMD)** patients and controls by generalizing **Multiscale Embedded Gene Co-Expression Network Analysis (MEGENA)**. We propose a curated set of module quality evaluation metrics to choose appropriate statistical distance-based or information-theoretic similarity measures over simple linear correlation when representing the similarities between genes.

Furthermore, we design and implement a stability test to ensure the robustness of the detected hub genes in the presence of noise, and propose differential module eigengene analysis to understand the up- and down-regulation of each module between the disease and control groups. Besides detecting robust hub genes and modules that are supported by prior findings, we also identify previously undiscovered hub genes that can potentially lead to further biomedical research into AMD disease mechanisms and the development of new treatments.

Motivation

- **Age-related Macular Degeneration (AMD)** is the leading cause of irreversible vision loss in adults over 50, affecting ~ 20 million people in the U.S. alone.
- AMD is driven by **coordinated gene networks**, not isolated single genes – arising from the combined effects of aging, genetics, and environment.
- Identifying individual disease genes must be supplemented with **clustering analysis** to reveal gene modules and their relationships.
- **Network-based analysis** is needed to go beyond single-gene approaches and uncover the regulatory structure underlying the disease.

Challenges

- High-dimensional RNA-Seq data with many variables and few samples.
- Noise and biological heterogeneity across patient samples reduce reproducibility.
- Simple **linear correlation** (Pearson) is insufficient to capture complex, nonlinear gene–gene relationships.

Contributions

- **Generalize MEGENA** (Multiscale Embedded Gene Co-Expression Network Analysis) beyond linear correlation.
- Use **information-theoretic similarity metrics** to weight gene–gene edges.
- Propose a curated set of **module quality metrics** to select the best similarity measure.
- Design **stability testing** to ensure hub genes are robust to noise.
- Apply **Differential Module Eigengene (DME)** analysis to detect up-/down-regulation between cohorts.

Dataset

Bulk RNA-Seq from human retinal tissue (controls + late-stage AMD donors).

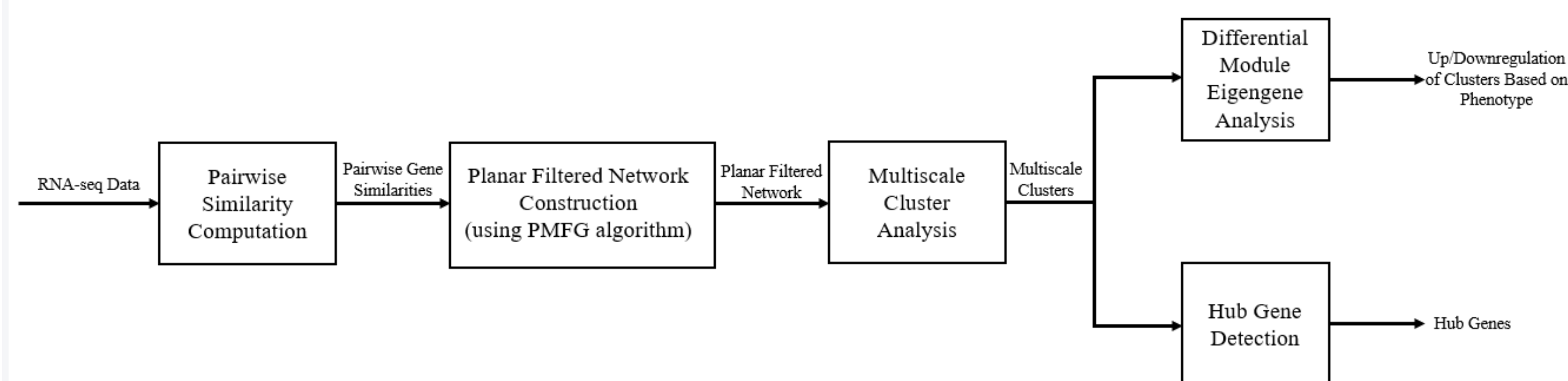
105
Control Samples

61
Late-Stage AMD

81
Curated Genes

A curated set of 81 genes selected by explainable ML, enriched for **immune, complement, and inflammatory** pathways.

Pipeline Overview



From pairwise gene similarity and PFN construction to multiscale clustering, hub-gene detection, and phenotype-specific DME analysis.

Methods

1. Similarity metrics. We replace linear correlation with metrics that capture nonlinear, distributional structure. Adjusted Mutual Information (AMI) was selected for this dataset:

$$\text{AMI}(X, Y) = \frac{\text{MI}(X, Y) - \mathbb{E}[\text{MI}(X, Y)]}{\max\{H(X), H(Y)\} - \mathbb{E}[\text{MI}(X, Y)]}$$

where MI is mutual information and $H(\cdot)$ is Shannon entropy. Jensen–Shannon divergence (JSD) and Pearson correlation were also compared.

2. Planar Filtered Network (PFN). A sparse graph preserving only the strongest gene relationships, built with the Fast PFN Construction (PMFG) algorithm to control the false discovery rate.

3. Multiscale Clustering. Detects large modules and nested submodules, then identifies **hub genes** – highly connected, central regulatory genes that are potential disease and therapeutic targets.

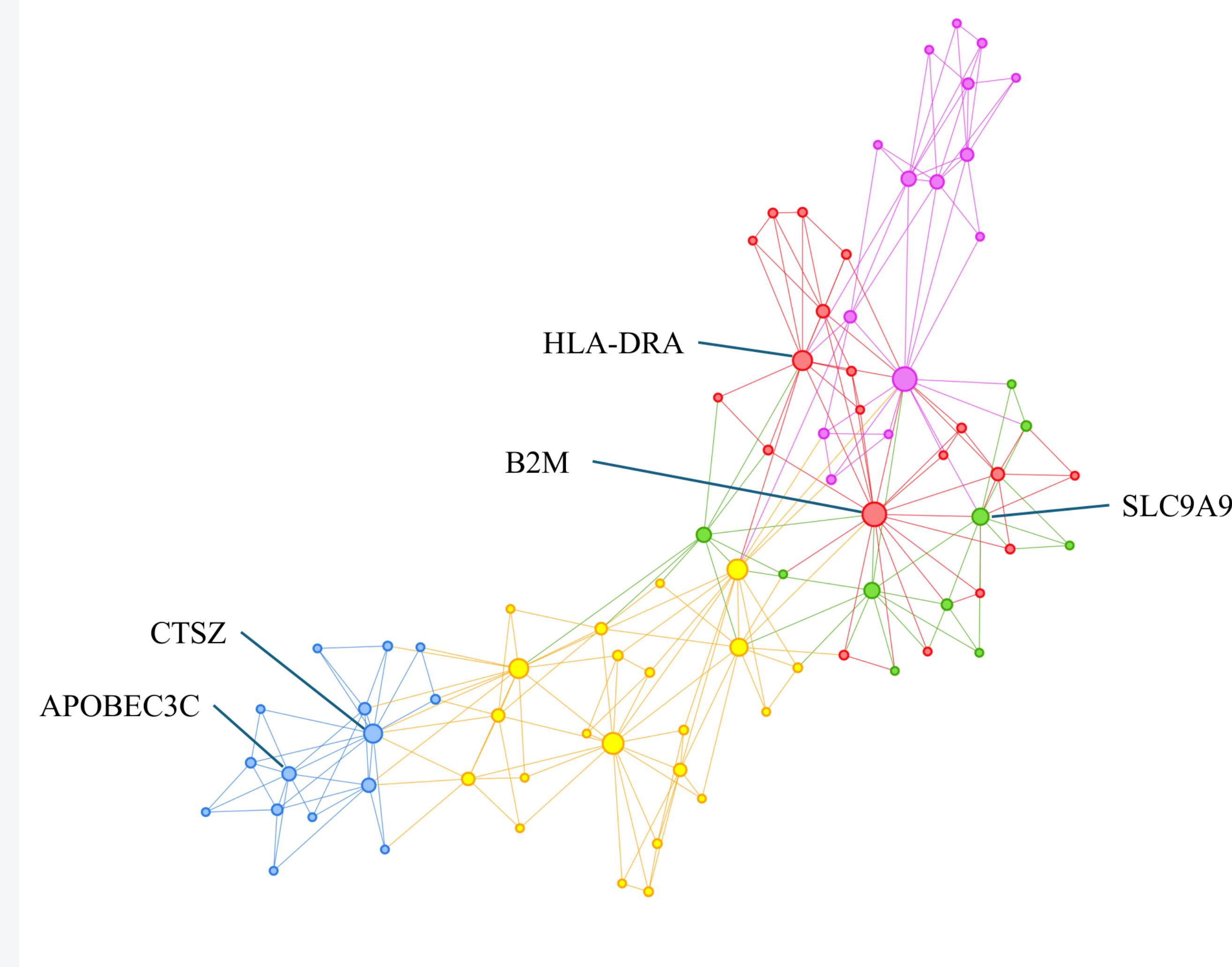
Module Quality Metrics

Five complementary graph metrics select the best similarity measure. **AMI** gave the most favorable balance of separation and coherence.

Metric	Modul.	Cond.	Dens.	Trans.	Corr.
Pearson	0.506	0.239	0.237	0.419	0.644
AMI	0.544	0.209	0.325	0.481	0.637
JSD	0.326	0.095	0.222	0.412	0.653

Higher modularity, density & transitivity and lower conductance indicate stronger, better-separated modules.

Results: Hub Genes & Modules



AMI-based generalized MEGENA network of the 81-gene set (combined cohort). Nodes colored by multiscale module; key hub genes labeled.

- Combined hubs: **SLC9A9**, **B2M**, **HLA-DRA**, **CTSZ**, **APOBEC3C**.
- **HLA-DRA** and **B2M** are hubs across *all three* cohorts.
- Shared immune/complement hubs (**C1R**, **SERPING1**) persist across control and disease cohorts.

Stability Testing

Each run appends **419 random genes** (total 500); repeated 100 $\times$. Hub recurrence (out of 100) confirms robustness to noise.

Gene	Combined	Control	Disease
C1R	99	94	93
HLA-DRA	84	70	81
SERPING1	84	43	94
B2M	66	64	46
CFB	–	–	65
CTSZ	17	–	59

C1R, **HLA-DRA**, **B2M**, and **SERPING1** remain the most frequently detected hubs – evidence of stable co-expression structure.

DME Analysis & Conclusion

DME tracks module up-/down-regulation between control and late AMD via the **module eigengene** (first principal component). All modules show statistically significant shifts ($p < 10^{-5}$, t -test & Wilcoxon).

- Nonlinear similarity metrics improve gene co-expression networks.
- Robust hub detection holds across patient cohorts and noise.
- Multiscale modeling reveals nested gene-module structure, reinforcing **immune dysregulation** as a core driver of AMD.

A robust, generalizable network framework for gene analysis in Age-related Macular Degeneration.